A Latent Oscillator Measurement Model for Simulating Emotion Dynamics in Video

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Abstract

Automated facial-expression scoring turns video into dense time series of model-derived visible-expression scores. These scores require method validation before researchers interpret dynamic structure as latent emotion dynamics. We introduce the Latent Oscillator Measurement Model (LOMM), a known-truth simulation framework for facial-expression-recognition-like video score series. LOMM generates latent oscillator trajectories, gates their activity over time, and maps them through block loadings into noisy classifier-output score channels with controlled dimensionality, loading strength, measurement error, feedback regime, and activity pattern. We calibrated the generator against MAFW-derived score panels using targets for amplitude dispersion, local movement, temporal persistence, directional-regime complexity, spectral complexity, and dimensional spread. The calibrated generator reproduced several local temporal and dimensional signatures, with a remaining amplitude-dispersion mismatch. We then compared Dynamic Exploratory Graph Analysis (DynEGA), GraphicalVAR, and bounded GIMME in known-truth recovery simulations. DynEGA showed the strongest dimensional recovery and completed the expanded grid. GraphicalVAR often recovered dimensionality but was slower and more sensitive to weak measurement conditions, consistent with partial-correlation attenuation of common-cause measurement blocks. GIMME served as a flexible feasibility-stress comparator and was computationally limited in this design. LOMM lets researchers test whether dynamic methods recover latent organization, tolerate measurement error, and scale to video-derived behavioral time series before substantive interpretation.

Introduction

Recent advancements in deep learning and computer vision have made it increasingly feasible to transform video frames into time series of facial-expression score channels (S. Li & Deng, 2022; Meng et al., 2017). In a

facial-expression recognition task, an algorithm maps each image frame or video segment to classifier-output scores, usually as category probabilities or continuous scores for labels such as happiness, anger, sadness, fear, surprise, disgust, or neutral expression. For behavioral-science applications, these tools create a new kind of intensive longitudinal record: high-frequency, model-derived measurements of visible affective expression. These records differ substantially from self-report time series collected through ecological momentary assessment (EMA; Tong et al., 2005) or experience sampling methods (ESM; Hoemann et al., 2020). Facial-expression recognition (FER) scores are dense outputs of measurement systems applied to video frames, making them promising for studying the temporal organization of emotional expression at scale. Their use also raises a methodological problem: dynamic models need to separate latent expressive processes from the observed score series used to measure them.

The substantive motivation comes from video-rich behavioral settings in which visible affect is part of the behavior being studied. Facial expressions are central to human communication (Barrett et al., 2019; Mehu & Scherer, 2015; Stewart et al., 2009), but many behavioral-science questions concern how these expressions unfold over time across frames. In psychotherapy tasks, autism assessments, collaborative work, public communication, and social-media videos, the relevant questions are dynamic: whether expressions return quickly after perturbation, whether affective displays settle into stable or volatile regimes, and whether many model-output score channels reflect fewer latent expressive dimensions (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017).

Automated facial-analysis systems now make this kind of work feasible at scale. Datasets such as FER2013 (Goodfellow et al., 2013) supported deep learning approaches to facial-expression recognition, especially Convolutional Neural Networks (Arriaga et al., 2017; CNNs; Ezerceli & Eskil, 2022). Most early systems inherited their emotion categories from the training data and focused on fixed classification tasks, commonly returning probabilities for seven categories: anger, disgust, fear, happiness, sadness, surprise, and neutral facial expression (Wang et al., 2022). More recently, transformer-based and CLIP-like models have enabled zero-shot classification with user-specified emotion labels (Foteinopoulou & Patras, 2023; H. Li et al., 2023; Tomašević et al., 2026), making it possible to represent each video frame as a vector of prompt-defined classifier scores. Applied studies are already using such automated scoring in clinical, collaborative, and social-media settings (Halamová et al., 2023; Holiday et al., 2023; Q. Li et al., 2024; Manelis-Baram et al., 2025).

This technical progress also creates a validation problem. FER-like score series are model-derived, multivariate, and error-prone measurements of visible behavior. Their interpretation depends on whether the observed score channels are treated as emotion variables in their own right, as indicators of social interac-

tion, or as noisy indicators of lower-dimensional latent affective processes (McKeown & Allison, 2024). For dynamic modeling, this distinction has practical consequences: measurement error can change the structure recovered by dynamic-network methods, making it important to separate latent temporal organization from the fallible score series used to observe it (Kankaanpää et al., 2026).

The aim of this paper is to introduce the Latent Oscillator Measurement Model (LOMM), a generative simulation model for FER-like score channels (Boker, 2015; Boker & Nesselroade, 2002; Ollero et al., 2023). LOMM first generates latent expressive trajectories from second-order oscillator dynamics and then maps those latent processes into FER-like classifier score outputs through an explicit measurement layer. LOMM is a known-truth data-generating model for simulation-based validation: it lets researchers control latent dimensionality, feedback regime, loading strength, measurement error, and activity pattern before testing whether dynamic methods recover the intended structure.

The paper contributes this model and evaluates its use in three linked steps, shown in Figure 1. First, we specify the LOMM data-generating and measurement components for FER-like video score panels. Second, we compare simulated panels with empirical MAFW-derived FER-like score panels to assess whether the generator reproduces selected temporal and dimensional signatures (Panels A–D). Third, we use the resulting known-truth simulations to evaluate dynamic dimensional recovery with DynEGA, GraphicalVAR, and bounded GIMME under controlled measurement and dynamical conditions (Panels E–H). We then close by discussing when video-derived score panels can support behavioral-science claims about latent affective dynamics.

Expressive Affective Dynamics as an Oscillator Measurement Model

Affective-dynamics research treats emotion as a time-varying process. Across common approaches, the relevant features include average intensity, variability, instability, inertia, transitions between states, and return after perturbation (Hollenstein, 2007; Houben et al., 2015; Kuppens & Verduyn, 2017). For video-derived expressive behavior, the same logic applies to visible affect. A short clip can contain low expressivity, brief expressive episodes, gradual recovery, persistent activation, or alternation between expressive regimes. Modeling expressive affect therefore requires a representation that can describe both the level of a latent expressive process and its tendency to persist, change direction, or return toward equilibrium.

An oscillator model is useful when latent variables are defined as theory-guided expressive dimensions that

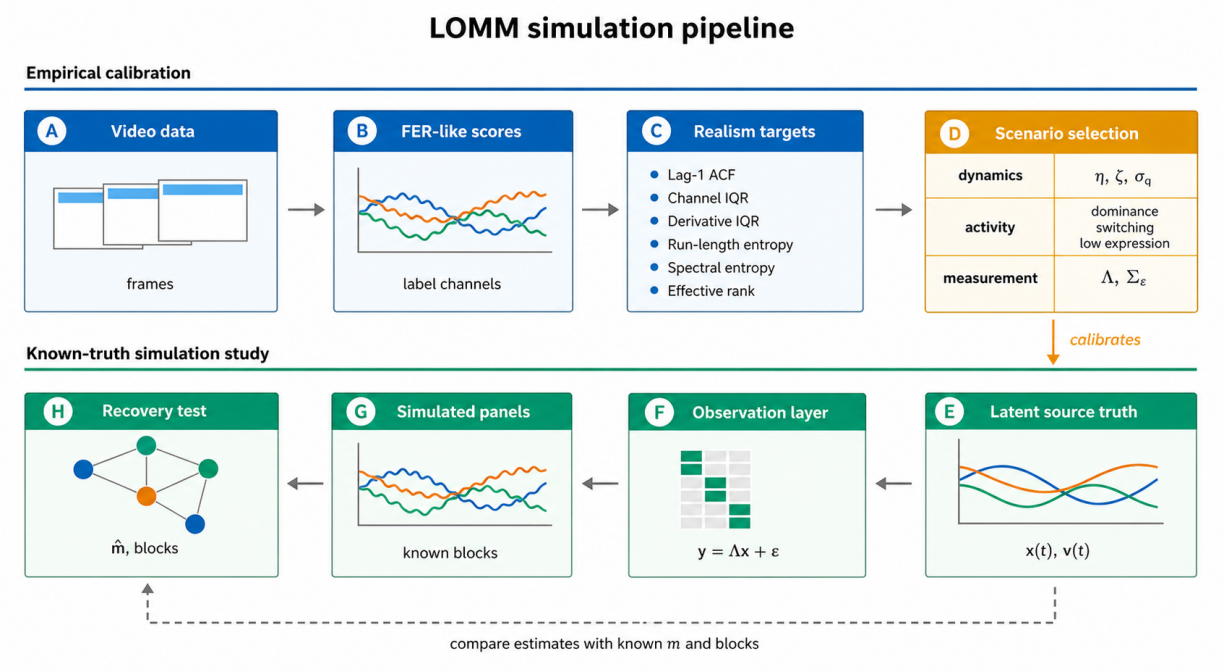


Figure 1: Visual overview of the LOMM simulation pipeline. Panels A–C show the empirical calibration step and realism targets for FER-like video score series. Panel D shows scenario selection through dynamics, activity, and measurement parameters. Panels E–H show the known-truth simulation, read from right to left: latent source-truth trajectories are passed through the observation layer to generate simulated score panels, and recovery tests compare estimated dimensions and blocks with the known latent structure.

organize FER model outputs. Affective structure can be specified in several defensible ways: core-affect and circumplex models organize affect around valence and arousal; PAD models add dominance or control; and Plutchik-style accounts arrange emotion families around opponent tendencies such as joy–sadness, trust–disgust, fear–anger, and anticipation–surprise (Plutchik, 2001; Russell, 1980, 2003; Russell & Mehrabian, 1977). Facial-expression research already uses this distinction: AffectNet includes discrete expression categories and continuous valence-arousal ratings, ABAW evaluates valence-arousal estimation alongside expression classification and action-unit detection, and integrated accounts of facial communication treat categories and dimensions as complementary descriptions (Kollias et al., 2023; Mehu & Scherer, 2015; Mollahosseini et al., 2019).

For a CLIP-like FER panel, this measurement logic can be made concrete. Frame-level prompt scores for happiness, amusement, trust, or smiling can serve as noisy indicators of an approach-affiliation trajectory. Anger, fear, disgust, contempt, or anxiety scores can indicate a threat-defense trajectory. Surprise, anticipation, confusion, or attention scores can indicate an orienting-novelty trajectory. LOMM generates these latent trajectories first, maps them through block loadings into classifier-output score channels, and gives the recovery method only the simulated score matrix. The known-truth question is therefore whether a

method such as DynEGA can recover the intended expressive blocks from CLIP-FER-like model-derived score channels. The model has two integrated parts: a latent oscillator process for expressive dynamics and a measurement model adapted from the direct autoregressive factor score model (H. Golino et al., 2022; Nesselroade et al., 2013).

Following Ollero et al. (2023), LOMM treats each latent expressive dimension as a continuous-time second-order state process. Let $i = 1, \dots, V$ index videos, t index time, and $f = 1, \dots, m$ index the latent expressive dimensions selected for a simulation. For video i, $x_{if}(t)$ is the latent level of dimension f, and $v_{if}(t) = \dot{x}_{if}(t)$ is its instantaneous rate of change. The implemented latent process is

$$\frac{\mathrm{d}}{\mathrm{d}t}\begin{bmatrix} x_{if}(t) \\ v_{if}(t) \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \eta_f & \zeta_f \end{bmatrix}\begin{bmatrix} x_{if}(t) \\ v_{if}(t) \end{bmatrix} + \begin{bmatrix} 0 \\ q_{if}(t) \end{bmatrix}, \qquad q_{if}(t) \sim \mathcal{N}(0, \sigma^2_{qf}), \tag{1}$$

or equivalently, $\ddot{x}_{if}(t) = \eta_f x_{if}(t) + \zeta_f \dot{x}_{if}(t) + q_{if}(t)$. This is an uncoupled multivariate system: η_f, ζ_f, and σ_{qf} may vary by factor, but the model used here does not include off-diagonal dynamic effects in which one latent expressive dimension directly changes the acceleration of another. We therefore interpret η_f as position feedback and ζ_f as velocity feedback. Under this sign convention, negative η_f pulls a displaced latent trajectory back toward its equilibrium level; stronger negative values imply stronger restoring acceleration. Negative ζ_f attenuates velocity over time, $\zeta_f = 0$ removes velocity feedback, and positive ζ_f can sustain or amplify ongoing change over the finite interval being simulated. The innovation $q_{if}(t)$ is a factor-specific shock process that perturbs the latent trajectory. In the simulations reported here, the continuous-time system is advanced with the exact linear transition matrix and initialized with $x_{if}(1) = 0.25$ and $v_{if}(1) = 2$.

For simulation-based evaluation, LOMM defines scenario conditions at two levels. First, the factor-specific parameters η_f, ζ_f, and σ_{qf} set the dynamical regime: negative velocity feedback produces damped return, zero velocity feedback removes damping, and positive velocity feedback can sustain or amplify change over the finite simulated interval. Second, time-varying activity weights determine which latent dimension dominates the generated score series. In single-regime scenarios, one dimension has higher activity across the full sequence; in switching scenarios, the active dimension changes after sampled dwell intervals; and in low-expression scenarios, all dimensions remain close to background activity. Formally, the trajectory passed to the measurement model is $\tilde{x}_{if}(t) = a_{if}(t)x_{if}(t)$, where $a_{if}(t)$ is a scenario-defined activity weight. This design lets the simulation vary damping, dominance, switching, and volatility while keeping the source-of-truth latent dimensionality and measurement blocks known.

The observable FER-like score vector is generated from the scenario-scaled latent trajectories through a block

measurement model. For video i, if each latent dimension has k observed indicators, then $p = mk$ and

$$\mathbf{y}_{it} = \Lambda_i \tilde{\mathbf{x}}_{it} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim \mathcal{N}(\mathbf{0}, \Sigma_i). \tag{2}$$

Each observed score loads on exactly one latent dimension: for indicator j in block B_f, $(\Lambda_i)_{jf} = \lambda + u_{ij}$ with $u_{ij} \sim \text{Unif}(-0.15, 0.15)$, and all off-block loadings are zero. The measurement-error covariance matrix is diagonal, $\Sigma_i = \text{diag}(\sigma^2_{\varepsilon,i1}, \ldots, \sigma^2_{\varepsilon,ip})$, with $\sigma^2_{\varepsilon,ij} = s^2_\varepsilon + r_{ij}$ and $r_{ij} \sim \text{Unif}(-0.01, 0.01)$. In the simulation grid, s_ε is reported as the measurement-error standard deviation, and s^2_ε enters the diagonal covariance matrix. Thus LOMM simulates measured expressive factors and their observed indicator blocks; neutrality is not part of the source-truth structure in the simulation.

For the empirical realism check, this linear block predictor is transformed to the observed score scale with a row-wise softmax link. If $\mathbf{z}_{it} = \Lambda_i \tilde{\mathbf{x}}_{it} + \varepsilon_{it}$ is the pre-link score vector, then the retained observation layer uses

$$y_{itj} = \frac{\exp(z_{itj})}{\sum_{\ell=1}^{p} \exp(z_{it\ell})},$$

equivalent to a softmax temperature of 1.00. No empirical log-mean channel intercept and no additional neutral or background denominator are added in the retained realism specification. The softmax link is used to put the empirical realism panels on a bounded, compositional score scale; the known-truth recovery target remains the latent block structure that generated the pre-link score channels.

As a concrete three-factor example, a Plutchik-informed specification could set $m = 3$ for affiliative/approach expression, threat/defensive expression, and orienting/novelty expression (Plutchik, 2001). FER-like indicators related to joy, trust, amusement, or smiling would form one block; indicators related to anger, fear, or disgust would form a second block; and surprise- or anticipation-related indicators would form a third. This specification illustrates the measurement logic of LOMM: the researcher states a theory-guided latent organization, the simulator generates known latent trajectories under that organization, and the classifier-output score channels are noisy indicators whose loading strength and measurement error are substantive simulation conditions (Kankaanpää et al., 2026). Alternative affective taxonomies can be encoded by changing the latent block map.

The same three-factor example is shown in Figure 2 to make the role of the latent shock term visible.

Figure 2 highlights the role of the innovation variance, σ^2_q, in LOMM. The reference panel shows the concep-

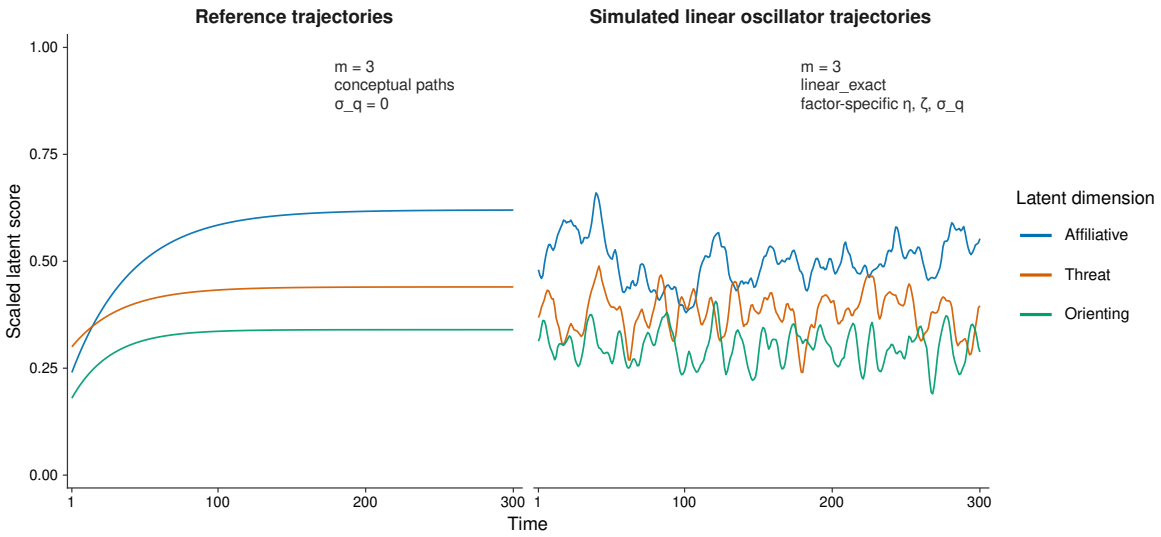


Figure 2: Three-factor LOMM example for Plutchik-informed latent expressive dimensions. The left panel shows smooth reference trajectories without dynamic shocks; the right panel shows trajectories generated by the implemented linear oscillator process with factor-specific position feedback, velocity feedback, and innovation variance.

tual latent ordering without dynamic shocks: affiliative expression rises quickly and settles above the threat and orienting trajectories. The simulated oscillator panel keeps this same structure, but nonzero factor-specific σ_q perturbs each latent process at each time point, producing local peaks, reversals, and irregular fluctuations around the underlying trajectory. After approximately time point 100, affiliative expression (blue) remains the dominant latent tendency, while threat (orange) fluctuates at a moderate level and orienting (green) stays lower with shorter local excursions. Thus, σ_q is the mechanism that turns a smooth conceptual model into latent score series that better resemble the uneven temporal structure of FER-like video data, without adding a neutrality trajectory or cross-factor coupling.

Method

The analyses follow the workflow in Figure 1. We first use empirical video scores to define realism targets for FER-like score series (Panels A–C). We then translate those targets into LOMM scenario conditions (Panel D) and evaluate dimensional-recovery methods on simulated panels for which the latent dimensional structure and measurement blocks are known (Panels E–H).

Empirical Calibration and Realism Targets

Figure 3 shows the empirical scoring target that motivates the simulation study. The figure uses one clip from MAFW, a dataset designed for dynamic facial expression recognition in the wild (Liu et al., 2022), whereas the realism check uses a 100-clip calibration sample scored with the same zero-shot FER-like procedure. We used the transforEmotion R package to generate frame-level classifier-output scores from the calibration videos (Tomašević et al., 2026). Each sampled video was decoded into frames, and each frame was scored against user-specified prompt channels with the package's `oai-large` vision model, which maps to OpenAI CLIP ViT-Large/14 (`openai/clip-vit-large-patch14`) (Radford et al., 2021). The scoring step returned one frame-level vector over the prompt-defined channels used here: anger, disgust, fear, happiness, neutral, sadness, surprise, contempt, anxiety, helplessness, and disappointment. Within each clip, short missing runs were interpolated, frames with complete usable scores were retained, and clipped score vectors were renormalized as probability-like outputs. These prompt-specific classifier outputs provide empirical targets for the LOMM observation layer through their temporal persistence, variability, and co-fluctuation.

The selected still frames from the same MAFW clip anchor the empirical example (Panel A). The corresponding score profiles illustrate the measurement problem (Panel B): different frames receive different scores across labels such as helplessness, fear, disgust, sadness, anxiety, and disappointment. The frame-level trajectories show that the highest mean channels change in bursts with irregular local movement (Panel C). The single-video DynEGA network applies Dynamic Exploratory Graph Analysis to the same clip's non-neutral score trajectories (Panel D). In this illustration, the method returns three communities among ten score channels, motivating the dimensional-recovery problem later studied under known truth.

The empirical calibration sample was assembled at the clip level from a label-diverse MAFW manifest and then retained after frame-score quality control. The retained panel contains 100 clips spanning the 11 MAFW single-label categories. After interpolation of short missing runs and complete-row filtering, usable clip lengths ranged from 17 to 50 frames, with a median of 36.5 usable frames. These retained frame-score panels define the empirical distributions used for the realism targets.

For the empirical realism check, empirical and simulated panels are summarized at two levels. Observable-channel metrics are computed directly from the FER-like score series within each clip. Community-score metrics are computed after applying pooled DynEGA to the score panel and extracting score summaries for the recovered communities. This distinction matters because a simulated panel can approximate the observable score channels while still producing a different lower-dimensional geometry, or conversely match the coarse latent organization while retaining imperfect channel-level amplitude behavior.

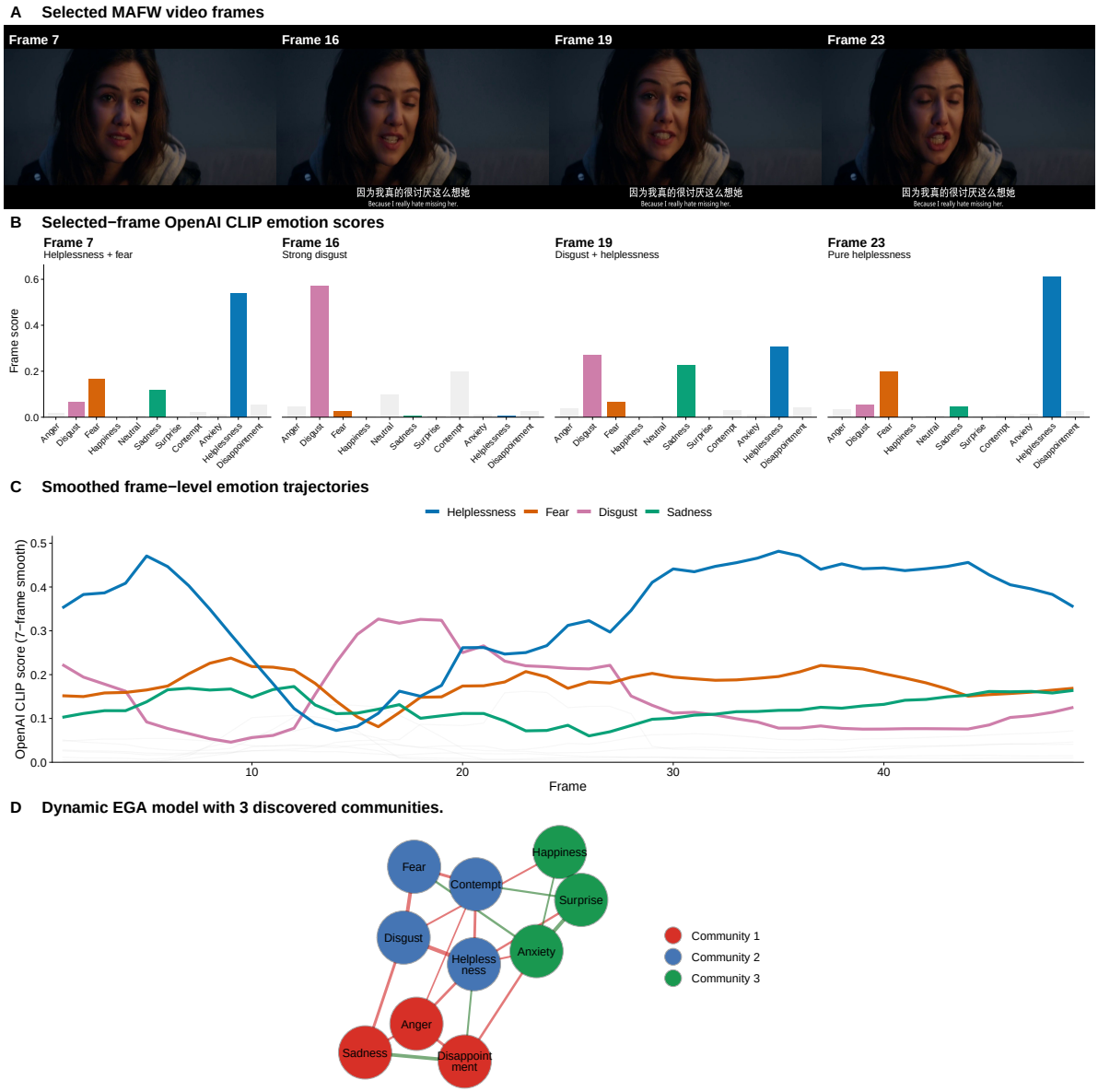


Figure 3: Single-video empirical illustration from the MAFW dataset. Panel A shows selected frames from one MAFW clip; Panel B shows zero-shot FER-like score profiles for the same frames; Panel C shows smoothed frame-level trajectories for the highest mean score channels; Panel D shows a Dynamic EGA network estimated from the same clip's non-neutral score trajectories.

The primary realism targets summarize dispersion, local movement, temporal structure, and dimensional spread. Channel IQR indexes score dispersion. Derivative IQR indexes the typical scale of frame-to-frame movement. Lag-1 autocorrelation indexes local persistence. Run-length entropy indexes directional-regime complexity. Spectral entropy indexes spectral complexity. Effective rank indexes multivariate dimensional spread.

For each metric, we compare empirical and simulated distributions using a scaled median difference and a scaled one-dimensional Wasserstein distance. The larger discrepancy determines the classification: Pass for scaled discrepancy ≤ 0.33, Partial for scaled discrepancy ≤ 1.00, and Fail for scaled discrepancy > 1.00. These criteria are simulation-construction checks; they do not validate the scoring model or the emotion labels.

Scenario Selection

Scenario selection links the empirical calibration targets to the simulation design. We group the manipulated conditions into three families (Panel D). The dynamics family varies the position feedback, velocity feedback, and innovation variance parameters, η_f, ζ_f, and σ_{qf}. The activity family varies whether one latent expressive dimension dominates across the full sequence, whether dominance switches after sampled dwell intervals, or whether all dimensions remain close to low-expression background activity. The measurement family varies the block loading matrix Λ_i and measurement-error covariance matrix Σ_i that generate the observed FER-like score panels.

The scenario set contains six activity-and-dynamics families. Four families represent generic expression regimes: low-expression single dominance, high-expression single dominance, high-expression switching, and high-expression rapid switching. Two families use the empirical calibration logic more directly: ranked-persistence and broad switching-geometry. Across these families, LOMM varies the latent dimensionality and measurement layer so that the number of observed channels is $p = mk$, where m is the number of latent dimensions and k is the number of observed indicators per dimension. The expanded DynEGA grid includes two-, three-, and four-dimensional source structures, and the bounded all-method comparator uses two- and three-dimensional source structures. The final crossed factors, replication counts, and method-specific grid sizes are specified in the dimensional-recovery simulation section below.

For each simulated video, LOMM first generates latent trajectories from the scenario-specific dynamic parameters. The activity schedule then scales those trajectories: non-switching scenarios keep one active dimension for the full sequence, rotating the dominant dimension across videos, whereas switching scenarios sample

dwell intervals and change the active dimension over time. The measurement layer is applied to the activity-scaled latent trajectories. Measurement error is reported as a standard deviation, with the corresponding variance entering the diagonal measurement-error covariance before small item-level variance jitter is added. The simulation design evaluates dimensional recovery under controlled combinations of dynamics, activity regimes, and measurement quality.

Dimensional-Recovery Simulation Study

The simulation begins with latent trajectories and the known number of latent expressive dimensions (Panel E). The LOMM observation layer maps each observed score channel to one latent dimension through the block loading matrix and adds item-level measurement error (Panel F), producing a simulated FER-like score panel (Panel G). The final step is the dimensional-recovery test, where the estimated number of dimensions and item memberships are compared against the known latent structure (Panel H).

We compare Dynamic Exploratory Graph Analysis [DynEGA; H. Golino et al. (2022)] with GraphicalVAR (Epskamp et al., 2018; Epskamp, 2024) and GIMME (Gates & Molenaar, 2012; Lane et al., 2019) because they represent adjacent but distinct approaches to intensive longitudinal structure. DynEGA is the focal dimensional-recovery method: it estimates groups of variables whose changes are coordinated over time. GraphicalVAR is a dynamic-network comparator that separates lagged temporal relations from contemporaneous partial associations. GIMME is an idiographic time-series framework that searches for relations that are common across individuals, shared by subgroups, or unique to individuals. In the present simulation, DynEGA tests whether a method designed for dimensional recovery can recover the known score blocks directly, GraphicalVAR tests whether a dynamic-network representation yields a similar partition after community detection, and GIMME tests whether a more flexible person-specific time-series framework is feasible under the same measurement conditions.

DynEGA combines time-delay embedding and derivative estimation from dynamical systems with exploratory graph analysis for dimensionality assessment (Boker et al., 2010; H. Golino et al., 2020; H. F. Golino & Epskamp, 2017). We apply DynEGA to the stacked video panel with video identity supplied as the grouping variable. The embedding dimension is selected by minimizing the Total Entropy Fit Index (TEFI) over candidate values (H. Golino et al., 2021), using $n_{\text{embed}} \in \{5, 10, 15, \lfloor T/4 \rfloor, \lfloor T/2 \rfloor\}$ where admissible. The estimated DynEGA population network is then partitioned into communities, which are interpreted as recovered groups of score channels. The expanded DynEGA grid uses Louvain community detection (Blondel et al., 2008) with the expand rule for unidimensional solutions. In the bounded all-method comparator, community detection is analyzed as a factor with Louvain, Leiden (Traag et al., 2019), and

Walktrap (Pons & Latapy, 2005), holding the unidimensional rule at expand.

GraphicalVAR estimates a graphical vector autoregressive model for intensive longitudinal data (Epskamp et al., 2018; Epskamp, 2024). In this framework, the temporal network contains directed lag-1 regression parameters, whereas the contemporaneous network contains partial correlations among residuals at the same time point after lagged effects have been taken into account. In multilevel applications, the model can also distinguish within-person temporal and contemporaneous structure from between-person associations among stationary means. We fit GraphicalVAR to the stacked simulated score panel and use the estimated contemporaneous partial-correlation matrix as the structural matrix for dimensional partitioning, because LOMM recovery targets an undirected grouping of score channels. The diagonal is ignored and the resulting weighted matrix is treated as an undirected graph. Positive edge weights define the graph, with absolute weights used only when no positive edges are available. Louvain, Leiden, and Walktrap are applied to the same fitted contemporaneous structure and are analyzed as the community-detection factor.

GIMME is an automated person-specific time-series modeling approach built around unified structural equation modeling and vector autoregressive representations (Gates & Molenaar, 2012; Lane et al., 2019). The algorithm searches for lagged and contemporaneous paths that improve fit across the sample, retains paths that are common to many individuals, and then allows subgroup-shared or individual-specific paths so that heterogeneous time-series structures do not have to be forced into one pooled model. In this simulation, each video is treated as the idiographic time series supplied to GIMME. We fit the VAR specification to the video-level score panels, extract the contemporaneous block from the person-specific Ψ matrices, and average those blocks across videos to obtain a common structural matrix for dimensional partitioning. The diagonal is ignored and the averaged matrix is treated as an undirected graph. As with GraphicalVAR, positive edge weights define the graph, with absolute weights used only when no positive edges are available. Louvain, Leiden, and Walktrap are applied to the same fitted GIMME structure and are analyzed as the community-detection factor.

All three methods are evaluated as dimensional-recovery estimators under known truth. DynEGA is evaluated on an expanded grid for scalability and on the bounded all-method comparator; GraphicalVAR and GIMME are evaluated on the bounded comparator because complete factorial estimation is computationally unstable in this setting. Community-detection algorithm is included as a diagnostic factor in the bounded comparator; method and simulation design remain the primary organizing variables.

Performance is summarized using dimensional hit rate, assignment accuracy for observed score channels, matched-score correlation, matched-score normalized root mean squared error (NRMSE), runtime, and fail-

ure or timeout accounting. Dimensional hit rate is the proportion of method-readout rows in which the method returns the correct number of latent dimensions. Assignment accuracy asks whether observed score channels are assigned to the correct latent blocks after optimal matching of estimated communities to true blocks. For score-based metrics, observed variables are standardized within the full panel and community scores are computed from the first principal component of each estimated block, with row means used when a principal-component score cannot be estimated. Matched-score correlation and matched-score NRMSE then compare these recovered score summaries with the known latent trajectories after sign-invariant matching between estimated communities and source dimensions. Runtime is recorded for each model fit. Non-converged fits, post-estimation extraction errors, and predefined timeouts remain in the denominators. For Graphical-VAR and GIMME, the same fitted structure is partitioned by multiple community-detection algorithms, so feasibility is also reported as unique structure-fit status counts.

The final simulation uses a split design. The expanded DynEGA grid varies the number of videos, series length, latent dimensionality, indicators per dimension, loading strength, and measurement-error standard deviation across 1,296 structural cells with 400 replications, yielding 518,400 DynEGA Louvain/expand method rows. The all-method comparator uses the bounded NVAR = 4 design with two- and three-dimensional source structures, two sample sizes, TimeP = 100, three loading strengths, three measurement-error standard deviations, and all six scenario families. This bounded comparator contains 216 structural cells with 100 replications. Louvain, Leiden, and Walktrap are included as community-detection levels, yielding 64,800 method-readout rows per method in the final comparator. Timeouts, failed fits, and post-estimation extraction errors are retained in the reporting denominator because they are part of method behavior in applied simulation workflows.

Results

Results are organized around four questions that follow the workflow in Figure 1. We first ask whether the calibrated LOMM generator produces FER-like score panels with empirical temporal and dimensional signatures. We then evaluate whether dimensional-recovery methods recover the known latent structure, examine how recovery changes with measurement quality, and report the computational feasibility of the compared methods.

Empirical Realism Results

Before evaluating dimensional-recovery methods, we asked whether the calibrated LOMM generator produced FER-like score panels that were sufficiently close to empirical panels for simulation construction. Figure 4 summarizes this check at two readout levels: the observed score channels and the lower-dimensional community scores extracted from the pooled DynEGA solution. The comparison evaluates selected distributional, temporal, spectral, and dimensional signatures of the empirical MAFW score panels, providing a simulation-construction check on the simulated panels.

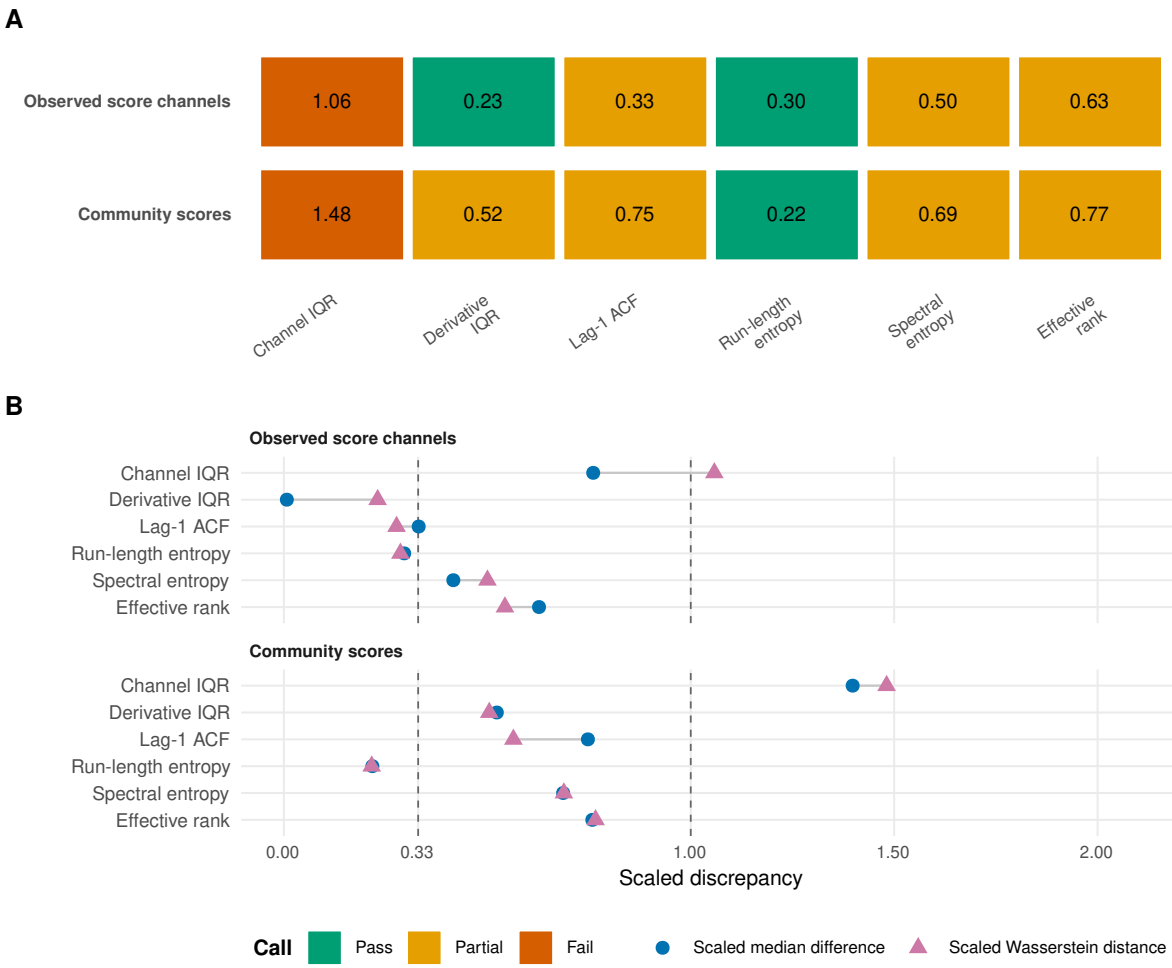


Figure 4: Empirical realism check for LOMM-generated FER-like panels. Panel A shows pass, partial, and fail classifications for the six realism targets at two readout levels: observed score channels and DynEGA community scores. Values are the larger of the scaled median difference and scaled Wasserstein distance. Panel B shows the two discrepancy components separately; dashed reference lines mark the pass threshold of 0.33 and fail threshold of 1.00.

At the observed-channel level, the softmax-linked generator passed the derivative-IQR and run-length entropy

checks. Lag-1 autocorrelation, spectral entropy, and effective rank were partial matches, indicating that the simulated panels were close to the empirical panels in local persistence, spectral complexity, and multivariate dimensional spread but not within the strictest threshold. The only failed observed-channel check was channel IQR, showing that the simulated observed channels remained more dispersed than the empirical score channels.

The community-score readout showed the same broad pattern with stronger amplitude mismatch. Run-length entropy passed, derivative IQR, lag-1 autocorrelation, spectral entropy, and effective rank were partial matches, and channel IQR failed. The retained Louvain community readout returned three communities with the same 6-3-2 channel-size pattern in the empirical and simulated panels, so the coarse dimensional organization matched in this readout even though the marginal amplitude of the community scores was too large. Taken together, these results support using the softmax-linked generator for dimensional-recovery simulation while keeping the amplitude and dispersion mismatch explicit as a limitation of the current calibration.

Simulation Results

The final recovery analysis separates scalability, direct method comparison, measurement-quality sensitivity, and computational feasibility. DynEGA was first evaluated on the expanded Louvain/expand grid, which contains 1,296 structural cells and 518,400 method rows. The direct method comparison then used the bounded NVAR = 4 design shared by DynEGA, GraphicalVAR, and GIMME. This bounded comparator contains 216 structural cells and 100 replications per cell. Louvain, Leiden, and Walktrap were included as community-detection readouts, yielding 64,800 method-readout rows per method, but the main comparison aggregates across these readouts and treats community detection as a diagnostic factor. Table 1 summarizes dimensional recovery, status counts, assignment accuracy as supporting block evidence, secondary matched-score diagnostics, and runtime by method.

Table 1: Overall method behavior in the final bounded NVAR = 4 comparator, aggregated across Louvain, Leiden, and Walktrap readouts. Hit rate is recovery of the known number of latent dimensions using all method-readout rows as the denominator; Assign. is optimal block-assignment accuracy. Score r and Score NRMSE are secondary matched-score diagnostics computed only for successful rows with recovered score summaries.

Method	Rows	ok	Timeouts	Errors	Hit (%)	Assign. (%)	Score r	NRMSE	Time (s)
DynEGA	64800	64800	0	0	95.3	96.0	0.725	0.649	2.61
GraphicalVAR	64800	62616	2184	0	80.3	90.4	0.717	0.659	86.10
GIMME	64800	45882	12342	6576	25.9	67.2	0.604	0.842	280.50

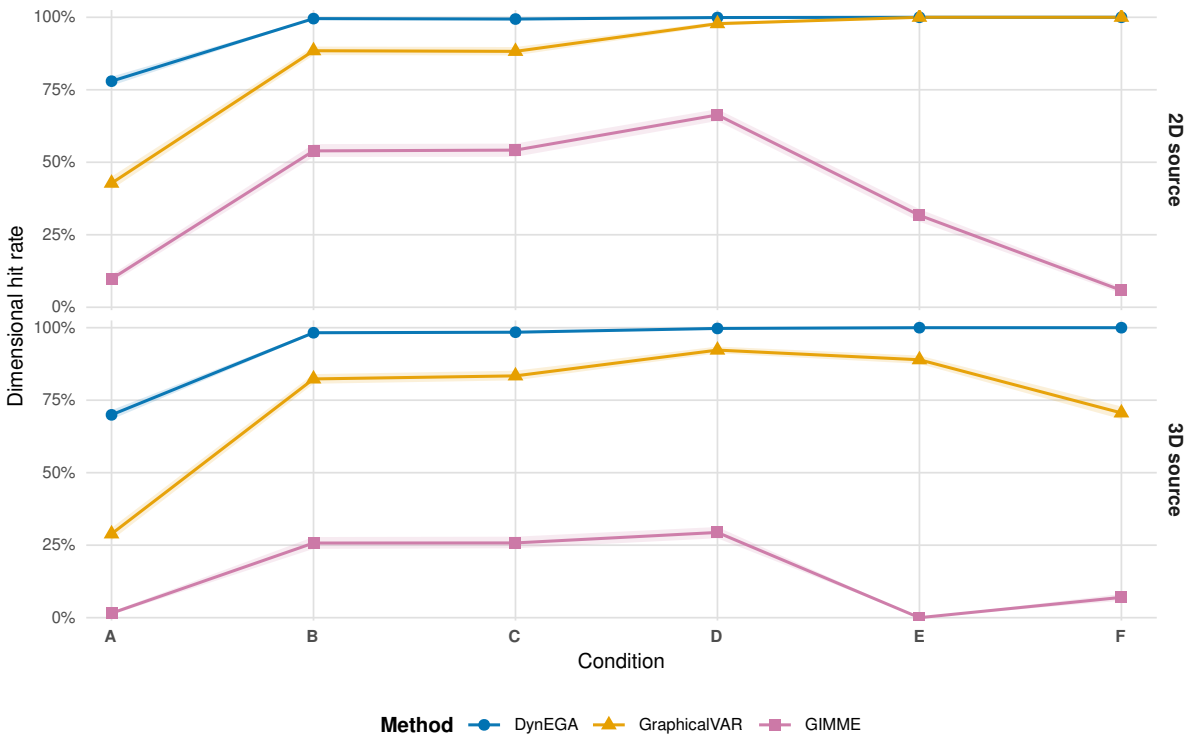


Figure 5: Dimensional hit rate across the final bounded NVAR = 4 method-comparison conditions. Lines show methods and shaded bands show cluster-bootstrap 95% confidence intervals, resampling generated panels while retaining their community-detection readouts. The condition codes are A = low-expression single-dominance, B = high-expression single-dominance, C = high-expression switching, D = high-expression rapid switching, E = ranked-persistence, and F = broad switching-geometry. Hit rates are status-inclusive and aggregated across Louvain, Leiden, and Walktrap readouts.

DynEGA was the only method evaluated on the expanded grid and completed all 518,400 Louvain/expand rows. On that grid, it recovered the correct number of dimensions in 93.7% of rows and assigned observed channels to the correct blocks with 95.2% accuracy. The bounded comparator in Table 1 then places DynEGA, GraphicalVAR, and GIMME on the same NVAR = 4 design. This comparison keeps unsuccessful fits in the denominator, so the reported hit rates combine structural recovery and estimator feasibility. Aggregating across the three community-detection readouts, DynEGA recovered the correct number of dimensions in 95.3% of bounded rows, GraphicalVAR in 80.3%, and GIMME in 25.9%. Table 1 supports this structural ranking with assignment accuracy: 96.0% for DynEGA, 90.4% for GraphicalVAR, and 67.2% for GIMME.

Figure 5 shows the same bounded-comparator results graphically. DynEGA retained the strongest recovery profile overall. The scenario-family pattern is also important: GraphicalVAR recovered the known dimensionality in many ranked-persistence and broad-switching-geometry conditions (E, F) when the structure fit completed, while recovery dropped in the low-expression single-dominance condition (A) and some

three-dimensional E and F fits reached the timeout boundary. GIMME showed the lowest status-inclusive recovery because many fits in the E and F families timed out or produced no usable contemporaneous structure for partitioning. The main simulation result is therefore a combination of target alignment, scenario sensitivity, and feasibility: DynEGA recovered the known block structure most consistently, GraphicalVAR was condition-sensitive and often recoverable, and GIMME was strongly feasibility-limited in this FER-like bounded design.

The factor screen focused the Results on dimensional hit rate. Method had the largest effect ($\eta_p^2 = .404$), followed by scenario family ($\eta_p^2 = .241$), measurement-error standard deviation ($\eta_p^2 = .146$), and loading strength ($\eta_p^2 = .036$). Community detection contributed essentially no variation ($\eta_p^2 < .001$). Because scenario differences are shown in Figure 5, the next comparison focuses on the two measurement-quality factors that directly alter the observed FER-like panels: loading strength and measurement error.

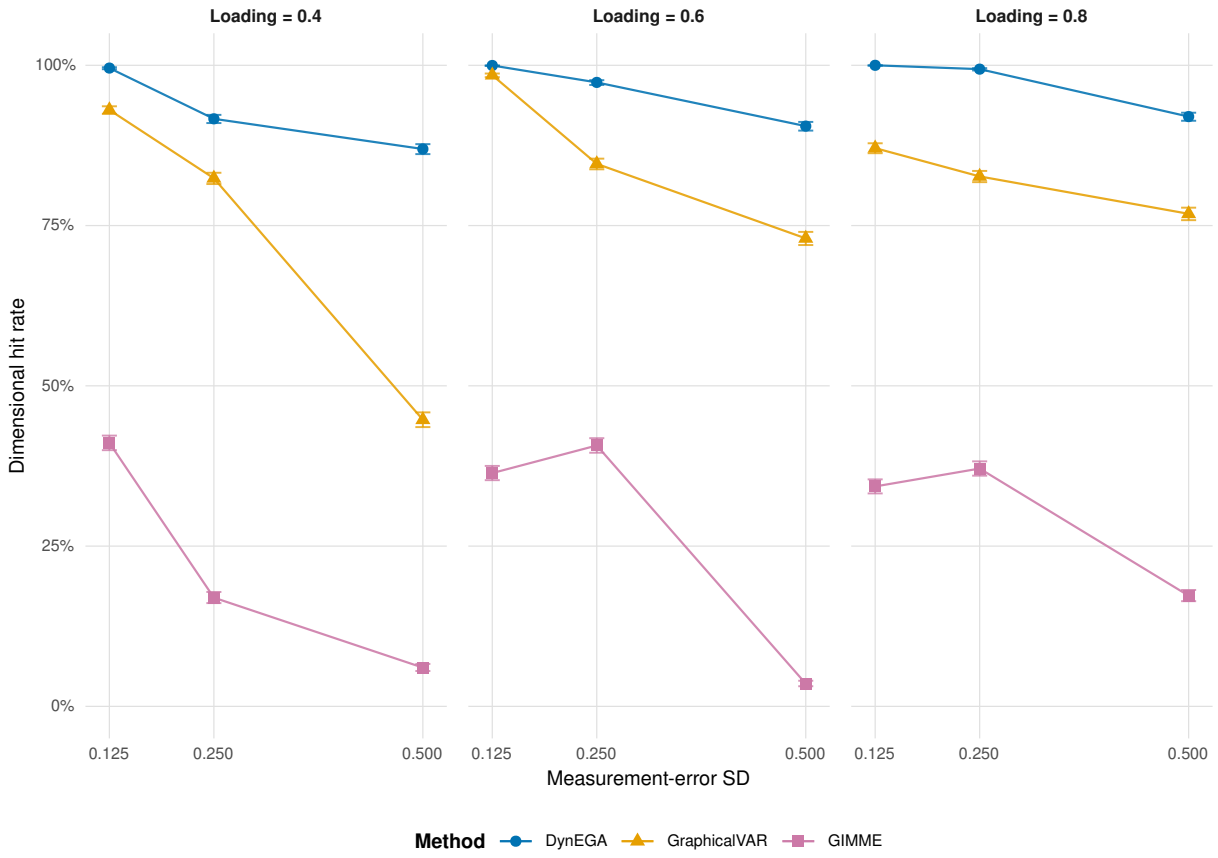


Figure 6: Dimensional hit rate stratified by measurement-error standard deviation and loading strength in the final bounded NVAR = 4 comparator. Columns show loading strength; lines and symbols show methods. Error bars are Wilson 95% confidence intervals computed from all method-readout rows in each stratum, aggregated across community-detection readouts.

Figure 6 shows how method differences changed across measurement quality. DynEGA was most stable across the loading-by-error grid, although low-signal scenarios still reduced recovery. GraphicalVAR was more sensitive to the measurement layer: its hit rate was weakest when loadings were low and measurement error was high, and improved when observed channels carried a stronger signal from their source dimensions. GIMME remained lower on status-inclusive hit rate across most strata because errors and timeouts occurred before dimensional recovery could be evaluated.

Table 2: Marginal score-quality diagnostics in the final bounded NVAR = 4 comparator. Rows use successful method-readout rows with recovered score summaries. Higher Score r and lower Score NRMSE indicate closer recovery of the known latent trajectories.

Factor	Level	Rows	Mean Score r	Mean Score NRMSE
Method	DynEGA	64800	0.725	0.649
Method	GraphicalVAR	62616	0.717	0.659
Method	GIMME	45882	0.604	0.842
Scenario family	A: low-expression single-dominance	32400	0.396	1.167
Scenario family	B: high-expression single-dominance	31791	0.634	0.972
Scenario family	C: high-expression switching	31800	0.581	0.894
Scenario family	D: high-expression rapid switching	30909	0.716	0.686
Scenario family	E: ranked-persistence	23910	0.985	0.118
Scenario family	F: broad switching-geometry	22488	0.999	0.034
Measurement-error SD	0.125	54771	0.814	0.561
Measurement-error SD	0.25	59181	0.704	0.694
Measurement-error SD	0.5	59346	0.562	0.845
Loading strength	0.4	59466	0.631	0.767
Loading strength	0.6	58428	0.703	0.686
Loading strength	0.8	55404	0.740	0.654

Table 2 reports mean matched-score diagnostics by the main design factors. These summaries describe score quality after a usable structure was estimated. Mean Score r was similar for DynEGA and GraphicalVAR and lower for GIMME; mean Score NRMSE followed the same ordering. Scenario family produced the largest differences: ranked-persistence and broad switching-geometry had high mean Score r and low mean Score NRMSE, whereas the low-expression single-dominance family had the weakest score recovery. Measurement quality also moved in the expected direction: higher measurement error reduced mean Score r and increased mean Score NRMSE, while stronger loadings improved both diagnostics. The accompanying factor screen showed the same hierarchy of influences: scenario family and measurement-error standard deviation explained the largest share of variation in both diagnostics, method and loading strength also explained meaningful variation, and community detection contributed essentially no variation. These score-quality results reinforce the measurement interpretation of the simulation and sit alongside the primary comparison

based on dimensional recovery and feasibility.

Computational Feasibility

Computational feasibility was evaluated separately from recovery accuracy because the methods differed in both recovery behavior and scalability across the simulation design. Figure 7 summarizes successful-fit runtime in the final bounded comparator, and the text below accounts for full-grid completion, timeouts, and post-estimation errors. This distinction is important for interpreting the comparator results: GraphicalVAR and GIMME are informative bounded comparisons with more limited scalability across the full LOMM grid.

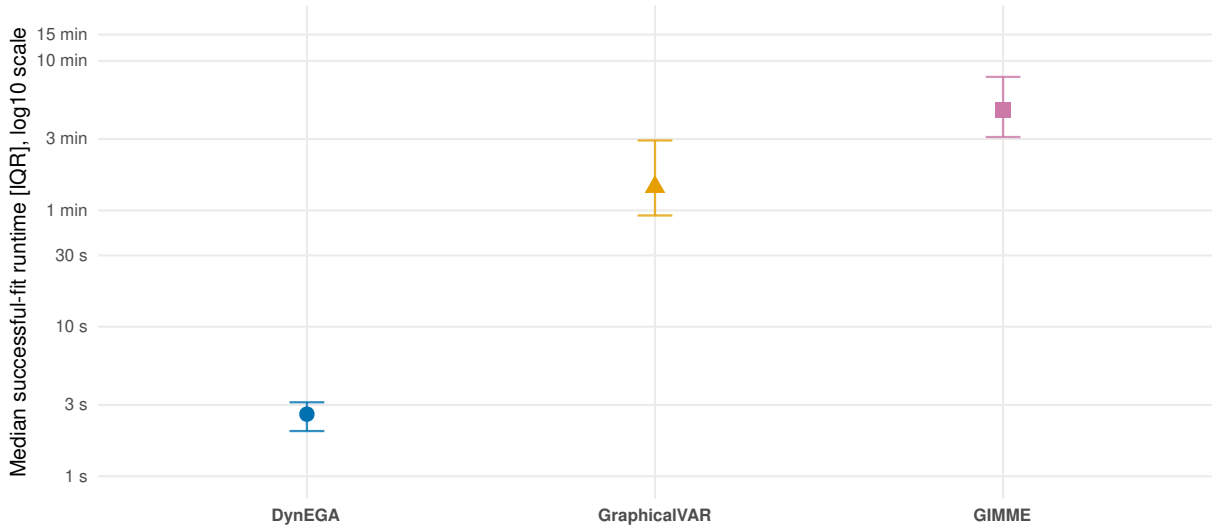


Figure 7: Runtime of the compared methods in the final bounded NVAR = 4 comparator. The y-axis is shown on a log10 time scale. Points show median runtime per successful method-readout row and intervals show the interquartile range, aggregated across community-detection readouts. For GraphicalVAR and GIMME, community-detection readouts share the same fitted structure within a panel.

DynEGA was substantially faster than the two model-based comparators in the bounded design. Median successful-fit runtime was 2.61 s for DynEGA, compared with 86.1 s for GraphicalVAR and 280.5 s for GIMME. The expanded DynEGA Louvain/expand grid also completed all 518,400 method rows without errors or timeouts, supporting its use for the broader scalability screen. GraphicalVAR completed 20,872 of 21,600 bounded NVAR = 4 structure fits and timed out on 728 fits, concentrated in the three-dimensional ranked-persistence and broad-switching-geometry conditions (E, F). GIMME completed 15,294 of 21,600 bounded structure fits, timed out on 4,114 fits, and returned post-estimation errors for 2,192 fits. Because GraphicalVAR and GIMME apply three community-detection algorithms to each fitted structure, these feasibility counts refer to unique structure fits before repeated CD readouts. Taken together, the Results support the abstract's method claim: DynEGA showed the strongest structural recovery and expanded-

grid scalability, GraphicalVAR provided informative bounded comparisons with sensitivity to measurement and scenario conditions, and GIMME was bounded by runtime, timeout, and extraction-error limits in this design.

Discussion

This study introduced LOMM as a known-truth framework for studying dimensional recovery in FER-like video score series. The strongest conceptual result is that LOMM turns dynamic dimensional recovery into a controlled perturbation problem. In the population design, m latent oscillator processes generate m measured score blocks. Within a block, observed channels share the same latent level and rate of change up to loading differences, activity weights, and measurement error. The recovery task is therefore to recover m rank-one dynamic subspaces from noisy, activity-gated, block-measured oscillator traces.

This framing clarifies why the method comparison is mainly about target alignment and scalability. DynEGA is designed to estimate groups of variables whose changes are coordinated over time, and this target is close to the population object generated by LOMM. In the ideal case, each score block forms a coherent dynamic subspace; finite samples, measurement error, weak loadings, activity switches, and oscillator amplitude differences perturb that block structure. DynEGA's high dimensional hit rate, high block-assignment accuracy, and complete expanded-grid coverage show that the estimated dynamic communities usually remained close enough to the known LOMM blocks for recovery, while the low-signal scenarios identify the boundary of that performance. The community-detection diagnostic did not materially change this conclusion because its effect on dimensional hit rate was negligible relative to method, scenario family, and measurement quality.

GraphicalVAR is a useful comparator because it estimates dynamic network structure, but its structural matrix is less directly aligned with a reflective block measurement model. The contemporaneous partial-correlation representation conditions away shared variance among indicators. Under a common-cause block structure, this can attenuate the within-block edges that define the latent measurement block, even when the marginal block signal is strong. This population alignment explains why finite-sample noise, lower loadings, higher measurement error, and temporal-model estimation can more easily disrupt the recovered partition. In the final bounded results, community-detection choice made very little difference once the GraphicalVAR structure fit had been obtained; the main limits were scenario sensitivity and three-dimensional timeouts in the ranked-persistence and broad-switching-geometry conditions (E, F).

GIMME addresses a broader problem than the one LOMM makes primary. It can represent group-level, subgroup-level, and individual-level time-series structure, which makes it attractive for idiographic dynamic

modeling. In this design, that flexibility came with weak status-inclusive dimensional hit rates, long runtimes, frequent timeouts, and post-estimation errors, concentrated in the ranked-persistence and broad-switching-geometry panels (E, F). GIMME therefore contributes mainly as a feasibility-stress comparator for this simulation.

The matched-score diagnostics add a narrower measurement-quality result. Among successful fits with recovered score summaries, scenario family and measurement-error standard deviation explained most variation in score correlation and Score NRMSE. This pattern shows that trajectory-level score recovery depends strongly on the signal structure of the simulated panel after a usable structure has been estimated. These diagnostics complement the structural-recovery and feasibility results.

The empirical realism check sets the boundary on this interpretation. The softmax-linked simulated panels matched the empirical FER-like score panels well for local movement scale and directional-regime texture, and they partially matched persistence, spectral complexity, and dimensional spread. Channel IQR failed at both the observed-channel and community-score readouts, showing that the current calibration still produces larger marginal amplitude dispersion than the empirical panels. This mismatch identifies the main calibration boundary: the current LOMM specification reproduces selected dynamic and dimensional signatures, and its bounded probability-like observation layer leaves marginal score dispersion too high relative to the empirical panels.

Several limitations follow directly from this design. FER-like scores remain model-derived observations of visible expression, so the realism check evaluates selected statistical signatures; label validity and exact empirical trajectories remain outside its scope. The present LOMM implementation uses independent oscillator dimensions, activity-gated block loadings, and diagonal measurement error; it does not claim to capture the full geometry of real affective systems. Future work should test richer measurement structures, bounded or compositional observation layers, additional empirical scoring systems, and empirical applications where known-truth simulation results guide substantive validation of video-derived affective dynamics.

Declarations

Funding

The authors received no specific funding for this work.

Competing interests

The authors declare that they have no competing interests.

Ethics approval

This study used simulations and secondary, publicly described reference materials. It did not involve new recruitment, intervention, or direct collection of human-participant data by the authors.

Consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The manuscript analyses use simulation outputs and derived FER-like score summaries generated for this study. The OSF archive will include the version-controlled manuscript repository, final simulation runners, figure-generation scripts, manuscript-facing derived outputs, provenance manifests, and from-scratch rerun instructions for the final simulation grids. Raw MAFW video materials are third-party dataset materials; access to the original videos is governed by the MAFW dataset providers. Derived empirical score summaries needed for the realism check will be included in the archive.

Code availability

All code needed to rerun the final LOMM simulations, rebuild the method-comparison summaries, regenerate the manuscript figures and tables, and render the manuscript will be available in the archived project repository. The repository includes the R simulation runners, empirical realism and method-comparison scripts, manuscript rendering source, claim ledger, and run manifests.

Authors' contributions

Aleksandar Tomasevic contributed conceptualization, formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Hudson Golino contributed formal analysis, methodology, resources, software, validation, visualization, writing the original draft, and reviewing and editing the manuscript. Alexander P. Christensen contributed methodology, software, writing the original draft, and reviewing and editing the manuscript.

Open Practices Statement

The study was not preregistered. The OSF archive will include analysis code, simulation code, manuscript source files, generated figure and table inputs, derived empirical score summaries needed for the realism check, and provenance materials for rerunning the final simulations from scratch. Raw MAFW video data are third-party materials governed by the dataset providers.

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Appendix A: Population Perturbation View of LOMM Recovery

This appendix formalizes the simplified LOMM population object behind the Discussion. The derivation concerns the population structure that any dynamic dimensional-recovery method receives in this simulation: observed score channels in the same latent block share coordinated changes because they are noisy indicators of the same activity-scaled oscillator. It does not characterize the fitted DynEGA algorithm, its community-detection implementation, or the exact behavior of GraphicalVAR or GIMME software.

A.1 Simplified Population LOMM

Consider a population LOMM with m latent dimensions, k observed indicators per dimension, and $p = mk$ observed score channels. Let $B_1, \dots, B_m$ denote the true indicator blocks. For indicator $j \in B_f$, assume the equal-loading, equal-error measurement model

$$Y_j(t) = \lambda Z_f(t) + \varepsilon_j(t),$$

where

$$Z_f(t) = a_f(t) X_f(t)$$

is the activity-scaled latent oscillator for dimension f, $\lambda > 0$, and $\varepsilon_j(t)$ has mean zero and variance $\psi = s_\varepsilon^2$. Measurement errors are independent across indicators and time, and the centered latent processes Z_f are mutually independent with finite variance. The implemented simulation allows loading jitter and item-level error jitter, but this symmetric case isolates the block geometry that defines the known-truth recovery target.

A.2 Dynamic Coordination Graph

Define the lag-h score change as

$$\Delta_h Y_j(t) = Y_j(t+h) - Y_j(t),$$

and define a population dynamic coordination graph by

$$W_{j\ell}^{\Delta} = \left|\mathrm{Cor}(\Delta_h Y_j, \Delta_h Y_\ell)\right|.$$

For two distinct indicators $j, \ell \in B_f$,

$$\Delta_h Y_j(t) = \lambda \Delta_h Z_f(t) + \Delta_h \varepsilon_j(t),$$

so

$$\mathrm{Cov}(\Delta_h Y_j, \Delta_h Y_\ell) = \lambda^2 \,\mathrm{Var}(\Delta_h Z_f)$$

and

$$\mathrm{Var}(\Delta_h Y_j) = \lambda^2 \,\mathrm{Var}(\Delta_h Z_f) + 2\psi.$$

Therefore the within-block dynamic edge is

$$w_f^{\Delta} = \frac{\theta_f^{\Delta}}{1 + \theta_f^{\Delta}}, \qquad \theta_f^{\Delta} = \frac{\lambda^2 \,\mathrm{Var}(\Delta_h Z_f)}{2\psi}.$$

For $j \in B_f$ and $\ell \in B_g$ with $f \neq g$, independence of the latent processes and measurement errors gives zero cross-block covariance. Thus the population dynamic coordination graph has block-clique form:

$$W^{\Delta} = \mathrm{blockdiag}\left(w_1^{\Delta}(J_k - I_k), \ldots, w_m^{\Delta}(J_k - I_k)\right).$$

This is the population object most directly aligned with dynamic dimensional recovery: variables in the same block share the same latent change signal, while variables in different blocks do not.

A.3 Partial-Correlation Graph

The GraphicalVAR comparator is converted into dimensional partitions by applying community detection to a contemporaneous partial-correlation matrix. To see how this representation changes the same block structure, consider the population covariance matrix inside block B_f. Let

$$S_f = \mathrm{Var}(Z_f), \qquad \theta_f = \frac{\lambda^2 S_f}{\psi}.$$

The block covariance matrix is

$$\Sigma_f = \lambda^2 S_f \mathbf{1}_k \mathbf{1}_k^\top + \psi I_k = \psi(I_k + \theta_f \mathbf{1}_k \mathbf{1}_k^\top).$$

Using the Sherman-Morrison identity,

$$\Sigma_f^{-1} = \psi^{-1}\left[I_k - \frac{\theta_f}{1 + k\theta_f}\mathbf{1}_k\mathbf{1}_k^\top\right].$$

The off-diagonal precision entry is

$$(\Sigma_f^{-1})_{j\ell} = -\psi^{-1}\frac{\theta_f}{1 + k\theta_f}, \qquad j \neq \ell,$$

and the diagonal precision entry is

$$(\Sigma_f^{-1})_{jj} = \psi^{-1}\frac{1 + (k-1)\theta_f}{1 + k\theta_f}.$$

The absolute partial correlation between two indicators in the same block is therefore

$$w_f^P = \frac{\theta_f}{1 + (k-1)\theta_f}.$$

Because the full covariance matrix is block diagonal under the simplified population model, cross-block partial correlations are zero. The GraphicalVAR-style population graph preserves the ideal block support,

$$W^P = \text{blockdiag}\left(w_1^P(J_k - I_k), \ldots, w_m^P(J_k - I_k)\right),$$

but compresses the within-block signal. The corresponding marginal common-source edge would be

$$w_f^M = \frac{\theta_f}{1 + \theta_f}.$$

For the same signal-to-noise ratio θ,

$$\frac{w^P(\theta)}{w^M(\theta)} = \frac{1 + \theta}{1 + (k-1)\theta}.$$

When $k > 2$ and $\theta > 0$, $w^P(\theta) < w^M(\theta)$. As $\theta \to \infty$, $w^P(\theta) \to 1/(k-1)$, whereas $w^M(\theta) \to 1$. Thus partial correlation preserves the ideal block support but attenuates the shared-source signal that defines a reflective measurement block.

A.4 Perturbation Margin

The preceding graphs are population graphs. Estimated graphs can be written as

$$\widehat{W} = W + E,$$

where E includes sampling error, derivative-estimation error, model-estimation error, finite-T and finite-N error, regularization, convergence behavior, and other implementation-specific perturbations. For a block-clique population graph

$$W = \text{blockdiag}\left(w_1(J_k - I_k), \ldots, w_m(J_k - I_k)\right),$$

let $w_{\min} = \min_f w_f$. Suppose every off-diagonal edge is estimated within δ of its population value:

$$\max_{j \neq \ell} |\widehat{W}_{j\ell} - W_{j\ell}| \leq \delta.$$

For any node $j \in B_f$, the estimated total connection to its true block is at least

$$(k-1)(w_f - \delta),$$

whereas its estimated total connection to any wrong block is at most

$$k\delta.$$

Thus each node has stronger total connection to its true block than to any wrong block whenever

$$(k-1)(w_{\min} - \delta) > k\delta,$$

or equivalently

$$\delta < \frac{k-1}{2k-1} w_{\min}.$$

This is the elementary perturbation statement. Recovery is favored when the estimation perturbation is small relative to the weakest within-block signal. Because the partial-correlation edge $w^P(\theta)$ is smaller than the marginal shared-source edge $w^M(\theta)$ for $k > 2$, the same perturbation has less margin under a partial-correlation representation.

A.5 Link to Oscillator Parameters

The signal-to-noise ratios above become LOMM-specific when $Z_f(t) = a_f(t)X_f(t)$ is tied to the latent oscillator. In the stable continuous-time case with constant activity a_f,

$$\ddot{X}_f(t) = \eta_f X_f(t) + \zeta_f \dot{X}_f(t) + q_f(t), \qquad \eta_f < 0,\ \zeta_f < 0.$$

Under the stationary linear approximation,

$$\mathrm{Var}(X_f) = \frac{\sigma_{qf}^2}{2\eta_f \zeta_f},$$

which is positive because both η_f and ζ_f are negative. Therefore

$$\mathrm{Var}(Z_f) = a_f^2 \,\mathrm{Var}(X_f) = a_f^2 \frac{\sigma_{qf}^2}{2\eta_f \zeta_f}.$$

The contemporaneous block signal is controlled by

$$\theta_f = \frac{\lambda^2 a_f^2 \sigma_{qf}^2}{2\eta_f \zeta_f \psi}.$$

For the dynamic coordination graph,

$$\mathrm{Var}(\Delta_h Z_f) = 2a_f^2 \,\mathrm{Var}(X_f)\{1 - \rho_f(h)\},$$

SO

$$\theta_f^{\Delta}(h) = \frac{\lambda^2 a_f^2 \operatorname{Var}(X_f)\{1 - \rho_f(h)\}}{\psi}.$$

In the underdamped case,

$$\rho_f(h) = e^{\zeta_f h/2}\left[\cos(\omega_f h) - \frac{\zeta_f}{2\omega_f}\sin(\omega_f h)\right], \qquad \omega_f = \sqrt{-\eta_f - \frac{\zeta_f^2}{4}}.$$

Thus the perturbation margin depends on the same quantities varied in the simulation design: loading strength λ, measurement-error variance $\psi = s_\varepsilon^2$, activity a_f, shock variance σ_{qf}^2, position feedback η_f, velocity feedback ζ_f, lag h, and number of indicators k.

A.6 Interpretation

This derivation explains the target-alignment claim in the Discussion. In the simplified population LOMM, the true dimensions are disconnected weighted cliques or, equivalently, rank-one dynamic subspaces. Finite samples and estimator behavior perturb that graph. Recovery is favored when the within-block signal remains large relative to the perturbation.

DynEGA is aligned with this target because it is used here to recover groups of variables whose changes are coordinated over time. GraphicalVAR remains a reasonable network comparator, but the contemporaneous partial-correlation representation conditions on other indicators in the same reflective block and therefore compresses the shared-source signal. GIMME addresses a broader idiographic time-series problem with group-level, subgroup-level, and individual-level paths, so the population theorem treats it as a flexible but computationally bounded feasibility comparator in this study.

Appendix B: Empirical Realism Numeric Summary

Table 3: Numeric summary for the empirical realism check. The main Results section reports these calls visually in Figure 4; this appendix table gives the corresponding medians and discrepancy components.

Readout	Metric	Empirical median	LOMM median	Scaled median diff.	Scaled Wasserstein	Max discrepancy	Status
Channels	Channel IQR	0.020	0.064	0.76	1.06	1.06	Fail
Channels	Derivative IQR	0.014	0.013	0.01	0.23	0.23	Pass
Channels	Lag-1 ACF	0.521	0.624	0.33	0.28	0.33	Partial
Channels	Run-length entropy	0.785	0.746	0.30	0.29	0.30	Pass
Channels	Spectral entropy	0.755	0.685	0.42	0.50	0.50	Partial
Channels	Effective rank	5.114	5.959	0.63	0.54	0.63	Partial
Community scores	Channel IQR	0.247	0.832	1.40	1.48	1.48	Fail
Community scores	Derivative IQR	0.159	0.294	0.52	0.50	0.52	Partial
Community scores	Lag-1 ACF	0.540	0.745	0.75	0.56	0.75	Partial
Community scores	Run-length entropy	0.785	0.756	0.22	0.22	0.22	Pass
Community scores	Spectral entropy	0.751	0.637	0.69	0.69	0.69	Partial
Community scores	Effective rank	2.371	2.803	0.76	0.77	0.77	Partial